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An open-source, 3D-printed auger powder doser designed with generative artificial intelligence[†]

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Many automated laboratories still weigh out powders by hand, because commercial powder-dosing robots cost tens to hundreds of thousands of dollars. We describe a single-channel powder doser built from 3D-printed parts and off-the-shelf electronics for under \$200, plus an analytical balance. A \$6 microcontroller turns, tilts, and taps a printed auger, a tube with a screw inside, until the balance reads the requested mass. The mechanical parts were first designed with generative artificial intelligence (AI) rather than drawn in conventional CAD software: the team made every design decision and reviewed, printed, and tested every part, while AI coding agents and a chat-driven CAD tool produced the geometry. Asking the AI for the whole assembly at once failed repeatedly; asking for one part at a time, against dimensioned drawings, worked better. The team later redrew the parts in conventional CAD, and the tested doser uses some of these, including the auger and its drive gear. We tested it on 13 powders: seven food-safe powders and six relevant to alloy discovery. With fixed settings, ten powders moved 10–231 mg per auger revolution and three did not. In closed loop, all 24 doses of 1 g on seven powders landed within $\pm 5\%$ of the target, and those of the three alloy-relevant powders within 1%. Doses of 50 and 200 mg were less reliable, because fast powders overshot and slow ones ran out of time; ongoing controller work targets this. The design files, firmware, data, and complete AI interaction history are public.

1 Introduction

Self-driving laboratories (SDLs) pair automated instruments with machine-learning planners that choose the next experiment, and they have sped up discovery in chemistry and materials science by an order of magnitude or more.^{1–4} Software for running them,^{5,6} methods for planning experiments,⁷ and robots for handling liquids are now mature. Handling solids is not: most SDLs either work only with liquids or rely on a person to weigh out powders.^{8,9} The gap matters most where the powders are the experiment. In alloy discovery by additive manufacturing (AM, the 3D printing of metal parts), each candidate composition is made by blending elemental or pre-alloyed powders,^{10–14} so weighing many different powders accurately and repeatably is the step that limits the whole workflow.

Commercial powder-dosing systems exist and work well, but their cost and size put them out of reach of most academic

laboratories. Benchtop gravimetric (weight-based) dispensers such as the METTLER TOLEDO XPR/CHRONECT line, Labman dosing stations, and MTT's AM powder dispensers cost from tens of thousands to several hundred thousand dollars, are built around pharmaceutical and quality-control work, and usually dose one vial at a time from proprietary dosing heads.^{15,16} Pharmaceutical manufacturing meters powders at much higher flow rates with loss-in-weight feeders, which weigh the hopper continuously as it empties,^{17–19} and laboratory studies have dosed milligram quantities with capillaries, vibrating chutes, and augers.^{20–23} None of these is available as an open design that a laboratory can print, build, and modify. Open-source hardware has shown that 3D-printed instruments can reach research-grade performance at 1–10% of the commercial price and keep improving through community contributions,^{24–27} and platforms such as Jubilee and low-cost SDL kits have brought that model to laboratory automation.^{28,29}

This paper describes the first module of such an instrument: a single-channel powder doser in which a 3D-printed auger delivers powder into a cup on an analytical balance, and a microcontroller turns, tilts, and taps the auger until the balance reads the requested mass (Fig. 1). Each channel is a self-contained printable unit, and later versions will dose several

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[†] Supplementary Information available: bill of materials, construction guide, auger and outlet cross-sections, bench noise record, closed-loop dose record, run inventory, and AI usage log. See DOI: 00.0000/00000000.

powders (Section 4). We report the design, a test campaign on 13 powders with very different flow behaviour, and closed-loop dosing results, including where the current design fails.

This study's second contribution concerns how the doser was designed. Its parts were first designed without conventional CAD software such as SolidWorks or Fusion 360. They were modelled by generative artificial intelligence instead: large language model (LLM) coding agents wrote each part as code in the project's public repository, and we also explored a chat-driven CAD tool, Zoo Design Studio. The division of labour was the same throughout. People decided what to build, wrote the specifications, eventually supplied fully dimensioned drawings, and reviewed, printed, and tested every part; the AI tools turned those decisions into geometry. Research on generative CAD has produced large datasets and models,^{30–33} and LLMs can now write CAD code (scripts that CAD software turns into solid models) from written descriptions,^{34–37} but published evaluations test benchmark shapes rather than the design of a working instrument from start to finish. Every prompt, generated part, review comment, and failure in this project is preserved in a public repository,* so the doser doubles as a documented case study of AI-assisted hardware design. Once the AI-designed doser worked, the team recreated its parts in Fusion 360 to obtain editable models for release. Those sessions were recorded too, so they are documented in the same way and give a rough comparison between the two routes. The doser used for the tests in this paper carries some of these recreated parts, including the auger and the stepper pinion that drives it (see Experimental). We describe the workflow that worked best, the ways the AI tools failed, and what the approach cost and gained compared with conventional design.

2 Results and discussion

2.1 The dosing module

Fig. 1a,b shows the module as designed and as built. The printed auger tube is also the powder reservoir; there is no separate hopper. Powder is poured in through the open upper end of the tube, which is then closed with a screw-on cap. The tube (25 mm outer diameter, 21 mm bore, 250 mm long) is printed as one piece. Along the third of its length nearest the outlet, a single-start screw (the flight, 10.4 mm pitch) winds around an 8 mm central core; the rest of the tube is plain and holds the powder (Fig. 1c). The whole piece turns together, like an Archimedes screw, rather than as a screw inside a fixed barrel. A NEMA-11 stepper motor turns it through a printed 20-tooth pinion and a 44-tooth gear printed on the outside of the tube, a 2.2:1 reduction; this auger and pinion are the Fusion 360 recreations described below. A push–pull solenoid taps the tube through a printed collar to help cohesive powders move. A small eccentric-rotating-mass (ERM) vibration motor was also planned, but its driver never worked (Section 2.4), so the doser as tested relies on rotation,

tilt, and tapping. The tube rides on a hinged plate that two servos tilt through a 2:1 gear. Because the hinge axis passes through the outlet, the dose lands in the same place at every angle (Fig. 1d): the tube can be parked horizontally, where nothing flows, and tipped up to 45°, its maximum tilt, to dispense.

An analytical balance (A&D HR-100A, 0.1 mg readability) weighs the collection cup and reports to a \$6 Raspberry Pi Pico W microcontroller. The microcontroller runs a three-phase dosing procedure: fast continuous rotation, then small rotations, then solenoid taps until the reading is within tolerance of the requested mass (Fig. 1e; see Experimental). The structural parts are printed in PLA, most of them with support material, and the remaining parts are commodity items. The bill of materials comes to under \$200 for a single channel, excluding the balance, which most laboratories already own (SI Table S1). That is roughly two orders of magnitude below a commercial gravimetric dosing head.

2.2 How the parts were designed

The mechanical design went through four versions in its first six weeks. Every step, then and since, is recorded in a design log in the repository, which now has 128 entries. The project began as a hand-sketched mechanical scoop (a “powder excavator”) meant to ride on an existing computer-controlled (CNC) gantry. An AI-assisted survey of small-scale dosing mechanisms^{20,21} produced a list of eight candidates, among them a tap sieve, a metering wheel, a capillary wiper, and an auger. The team chose the auger for its dose resolution, its printability, and its tolerance of different powders, in line with its wide use in commercial micro-feeders.^{15,16} Before the second version, the team also printed four outlet geometries for the auger (SI Fig. S1). The second version packaged the auger, frame, drive, and actuators as a self-contained channel, the repeating unit of a future multi-powder doser. The third version changed the method rather than the mechanism. Instead of asking the AI tools for the whole module at once, the team specified each part separately (auger, geared auger, bracket, tap collar, mounting plate, baseplate, and servo hinge); AI tools modelled each part, and the team reviewed and printed it. The fourth version assembled these parts on the hinged baseplate with servo tilt. After this AI-designed doser worked, the team recreated its parts in Fusion 360 and changed some of them. The tested auger is filled through a capped end rather than through slots in the tube wall, and its flight no longer runs the full length of the tube; the tap collar was also redesigned to hold a different solenoid.

The specifications given to the AI grew steadily more detailed. The first requests described what the whole system had to do. Later requests described what each part had to do and how it met its neighbours: bolt patterns, shaft diameters, and mating faces. After that, the team supplied annotated engineering drawings, several of them fully dimensioned with the geometric relations between features. By the end, the AI's job had narrowed to modelling, that is, turning drawings into parametric CAD and working out derived dimensions and clearances. Every design decision stayed with the team.

*All issue threads, pull requests, and design-log entries cited in this paper are preserved at <https://github.com/vertical-cloud-lab/powder-doser>; “the repository” refers to this address throughout.

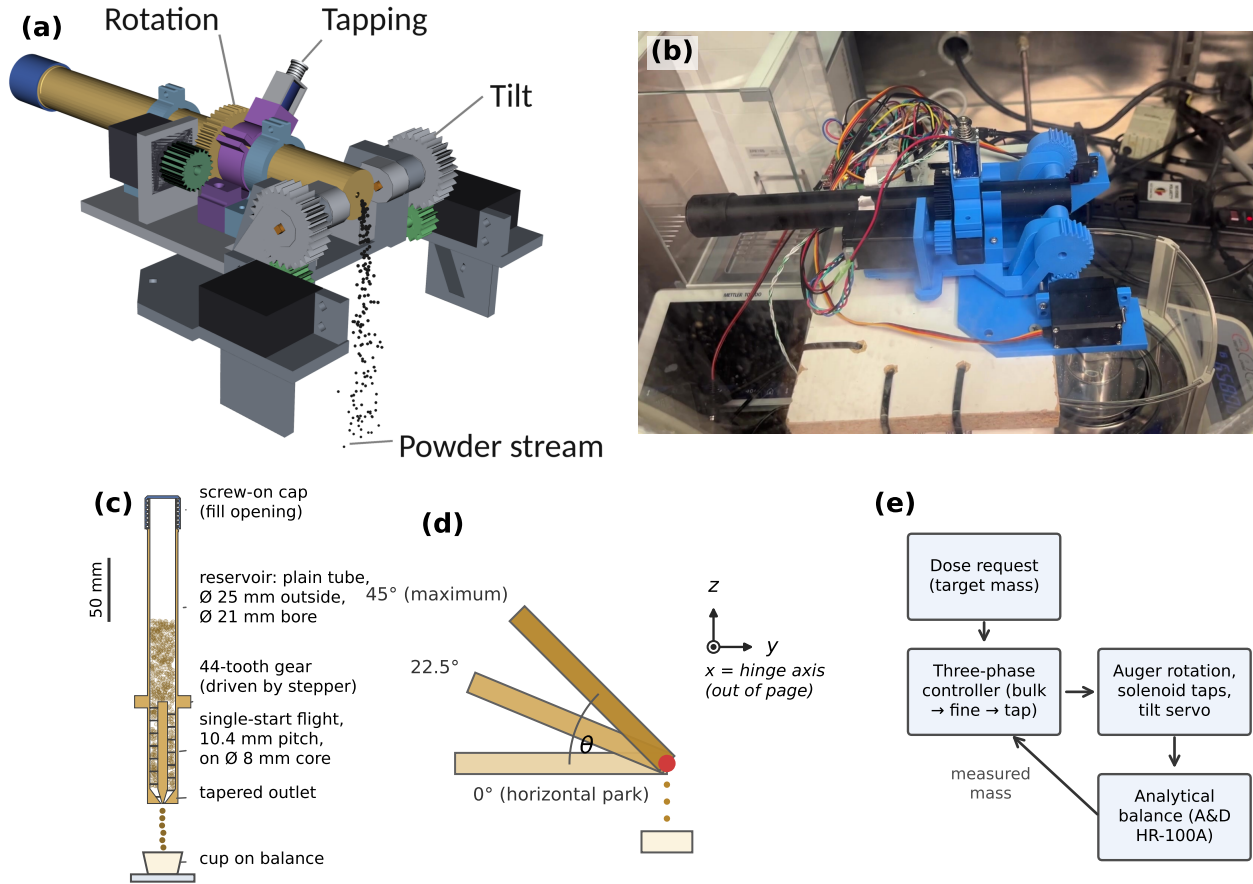


Fig. 1 The single-channel powder doser. (a) Annotated CAD model of the current design, showing the three motions used in this work: auger rotation, tapping, and tilt. The auger, cap, and pinion are the team's Fusion 360 parts; the tap collar and solenoid are modelled from photographs, and the plates and brackets are the earlier AI-modelled parts. (b) The module as built, inside a glove box, photographed from the same viewpoint during its first automated dispense (sodium chloride into a crucible on the balance). (c) Section through the tested auger and its screw-on cap, cut along the tube axis and drawn upright (outlet at the bottom). The tube is itself the reservoir and is filled through the capped end; the flight covers only the third of the tube nearest the outlet, and the 44-tooth gear is printed on the outside. Tube, flight, core, and gear are one printed piece and turn together. (d) Tilt about the fixed dispense point (red): the hinge axis x passes through the outlet, so the dose lands in the same place as the tilt angle θ changes from 0° (horizontal park) to 45° (maximum). (e) Closed-loop dosing: the three-phase controller (bulk, then fine, then tap) compares the requested mass with the balance reading and drives the actuators until the two agree. Humans made every design decision and reviewed, printed, and tested every part; AI tools produced the original CAD models without conventional CAD software, and the team later recreated some parts, including the auger, in Fusion 360.

Most parts were produced by LLM coding agents (the GitHub Copilot coding agent, mainly running Claude-family models) working directly in the repository. Given a task written as a GitHub issue, the agent wrote the part as parametric code (Python with build123d or CadQuery, or OpenSCAD). Each time the code changed, automated checks (continuous integration) rendered views of the part and exported printable STL mesh files. The agent then opened a pull request, and the team reviewed the part from the rendered images. We also explored two commercial text-to-CAD tools on the same parts: CADSmith, and Zoo Design Studio, which pairs a code-based CAD engine, scripted in the KittyCAD Language (KCL), with a conversational design agent called Zookeeper. In Zoo Design Studio, too, parts were built mainly by chatting with the agent rather than by editing the model by hand. Every prompt, generated file, render, and review comment is preserved in the repository’s pull requests and in a dedicated discussion thread.

2.2.0.1 One part at a time. Early attempts asked the agent for the complete module from a description of its layout. The results looked plausible and did not work. The first tap collar merged several functions into one body, with overlapping features, wrong tolerances, and no room for the actuators it existed to carry (Fig. 2a). An early bracket arrived with its two mounting plates unconnected. Requests at the level of the assembly repeatedly produced overlapping solids, floating parts, and changes nobody had asked for, and fixing one defect often introduced another (Fig. 2d). Restricting each request to a single part, with its interfaces listed explicitly, turned the agent into a productive collaborator (Fig. 2c). This fits the known weakness of current LLMs at reasoning about how shapes fit together in space,³⁴ and it suggests that splitting a design into parts with defined interfaces, standard practice for human engineers, is currently required rather than optional when AI does the modelling. Table 1 lists the failure modes we met and what caught each one. Each design-log entry records what triggered the change, the reasoning, and the outcome (built and worked, built with issues, failed, or superseded); a quantitative analysis of the log is planned for the multi-doser follow-up.

2.2.0.2 Checking the output. Most of the effort went into checking. Because any iteration could change geometry outside the part being edited, every generated part had to be re-inspected in full. In one case the agent produced four miniature test nozzles and swapped the labels on two of them, which would have invalidated a dispensing comparison had a person not noticed. Automated checks (that a part is a single solid, that its surface has no holes, and that key interface dimensions are correct) caught a useful share of defects cheaply. Small errors still escaped both the checks and visual review: a height mismatch of one print layer was found only when the part peeled off the print bed. One pattern held across the project. AI-generated alternatives were valuable early, while the design was still open, but late in the project the agent could not tell how close a part was to finished and kept changing features that were already right. AI-modelled parts therefore tended to stop at “good enough”, because each further round of edits cost time and risked breaking something

Table 1 Failure modes of the AI modelling tools observed in this project, and what caught or prevented each one.

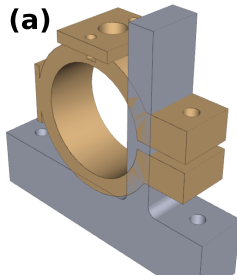
Failure	Example	Caught or prevented by
Parts that do not fit together	Whole-module prompts; first tap collar; bracket with unconnected plates (Fig. 2a,d)	Asking for one part at a time, with interfaces listed
Unrequested changes	Fixing one defect altered an unrelated feature	Full re-inspection of rendered views at every iteration
Mislabelled variants	Two of four test nozzles swapped relative to their documentation	Human cross-check before printing
Sub-millimetre errors	Height mismatch of one print layer	Only the failed print
Designing around bad inputs	Out-of-date neighbouring parts led to a hole, platforms, then a gap (Fig. 2e–h)	Supplying the correct upstream files
No sense of “finished”	Late iterations disturbed features that were already right	The team freezing the geometry
Wrong orientation	Solenoid mis-oriented on the tap collar over several coding-agent iterations	Switching the part to Zoo Design Studio

that worked.

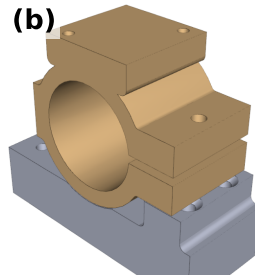
2.2.0.3 Comparing the tools. The coding-agent workflow was fully traceable and needed nothing beyond a web browser, but each iteration took minutes and every change had to be reviewed. CADSmith, given the same interface specifications, performed about as well as the coding agent. Zoo Design Studio was the team’s clear favourite. Its Zookeeper agent reasoned noticeably better about space, ran its own checks (such as counting solid bodies) before returning a result, and worked inside a live CAD view, so the team could inspect an editable model as it was built. Each iteration was slower, of the order of an hour, but one or two Zoo iterations achieved what took seven or eight with the coding agent. The final tap collar reached a usable design in three Zoo iterations after the same number of coding-agent iterations had failed to orient the solenoid correctly. At the time of testing, Zoo’s limits mirrored its strengths: imported reference parts could not be edited, there were no constraints between parts in an assembly, and manual edits were largely limited to changing parameter values, so the work stayed conversational. These rankings are the team’s impressions from one project, not measurements; a controlled comparison with identical prompts and parts is planned. The team also agreed that every AI route was slower, and gave lower-quality geometry, than experienced mechanical engineers working in conventional CAD software would have. A rough check came later, when two team members recreated the parts in Fusion 360 to produce editable models for release, changing some of them along the way. The narrated recordings for nine parts total about three hours of edited video,[†] roughly the length

[†]Playlist “Human redesign of AI-made parts”, https://www.youtube.com/playlist?list=PLZTWCFxzhTQv42fTuW2Tjs9o1KuWA_b-I (repository issue #120).

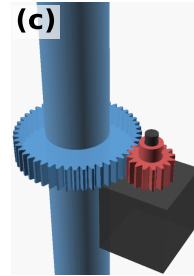
Tap collar, first AI proposal:
interferences, bad tolerancing,
no component clearance



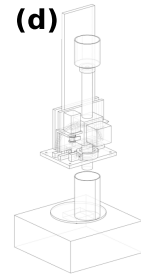
Tap collar after review iterations
(final part redesigned in Zoo)



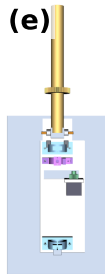
Geared auger + pinion
(part-by-part workflow)



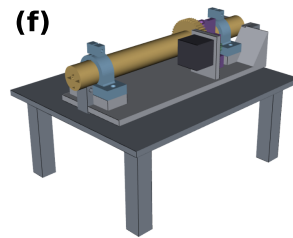
Whole-assembly attempt
(single prompt)



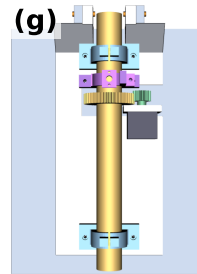
Iter. 1: unexplained
hole under gear



Iter. 2: raised platforms
instead of hole



Iter. 3: gap appears;
motor plate floats



Iter. 4: correct inputs
→ clean plate

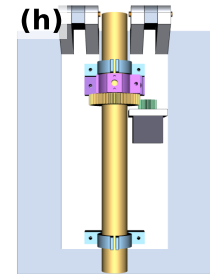


Fig. 2 AI-generated CAD, good and bad. (a) The first AI-generated tap collar: several functions merged into one body, with features overlapping one another, wrong tolerances, no room for the solenoid and vibration motor it had to carry, and no workable spatial layout. It could not have been built. (b) The same part after iterative review with the coding agent; the production collar was later remodelled in Zoo Design Studio, which reached a usable part in three iterations. (c) Geared auger and stepper pinion from the part-by-part workflow. (d) An attempt to generate the whole module from a single prompt: plausible at a glance, but with overlapping solids and floating parts that took longer to diagnose than the part-by-part route took to build. (e–h) Four review iterations of the mounting plate. The agent added an unexplained hole (e), then raised platforms (f), then a gap that left the motor plate floating (g), and produced a clean plate (h) only once the correct versions of the neighbouring parts were supplied. The cause was out-of-date input files, a human error that the agent silently designed around. All parts shown were modelled by AI tools to human-set requirements and reviewed by the team at each iteration; no conventional CAD software was used.

of a single Zoo Design Studio session. The recreations were not exact copies (the drive gears, for example, went from 16:48 to 20:44 teeth), and it is the recreated auger and pinion that turned during every test reported here. What the AI route bought instead was documentation as a by-product of the work. Every design intent, constraint, and revision exists as text, which makes the design reproducible, its history traceable, and the project easy to hand over, in a way that conventional CAD files are not.

2.3 How much powder one turn delivers

We tested the doser on 13 powders (Table 2). Seven are food-safe surrogates, chosen to cover a wide range of flow behaviour cheaply and safely: table salt (NaCl, the control), calcium lactate, xanthan gum, carboxymethyl cellulose (CMC), sodium alginate, and white and brown rice flour. The other six are relevant to the alloy-discovery work the doser is meant to serve: a gas-atomized AlSi10Mg alloy powder (near-spherical particles made by spraying molten metal), a coarse and a fine grade of silicon, sodium sulfate, barium chloride, and fumed silica, a very light flow aid. Each powder went through the same fixed sequence of test protocols with frozen settings (Table 3), so that differences between powders reflect the powders rather than the settings. In the first test round (4–21 August 2026), the food-safe powders were tested on an open laboratory bench and the other powders in a shared fume hood. The balance became much noisier from 20 August: with nothing moving, eight successive readings varied by 0.0 mg on every run up to 12 August and by up to 88 mg afterwards (SI Fig. S2). Bench conditions are therefore confounded with powder group, because every research-relevant powder was measured in the noisier period.

With one set of settings, the mass delivered per auger revolution at a 22.5° tilt varied 24-fold across the ten powders that moved, from 10 mg for sodium alginate to 231 mg for AlSi10Mg (Fig. 3a). Seven powders (AlSi10Mg, coarse silicon, sodium sulfate, calcium lactate, barium chloride, xanthan gum, and salt) delivered 146–231 mg per revolution, and CMC, white rice flour, and sodium alginate delivered 10–26 mg. Three powders did not move at any tilt: brown rice flour, fine silicon, and fumed silica. The powder mattered far more than the settings. Of the variation in mass per revolution across all measured revolutions (on a log scale), 70% was explained by which powder was loaded, 20% by tilt, 4% by the combination of the two, and 6% was scatter from one revolution to the next.

Delivery is not smooth. Balance readings streamed during continuous rotation rise in steps (Fig. 3b) that recur about every 4 s at 15 rpm, once per auger revolution; across ten powders, the median period from autocorrelation was 3.94 s against a revolution time of 4.00 s. This is what a single-start flight should do: each revolution carries one pocket of powder over the outlet. The pulsing limits how finely rotation alone can meter powder, and it is why the final approach to a requested mass uses taps (Section 2.5).

Plotting each powder's throughput against its variation from one revolution to the next (Fig. 3c) separates the powders into a readily dosed group, which delivers 150–230 mg per revolution

with 2–18% variation, and a slow group that delivers 10–26 mg. Both numbers come from about two minutes of bench time per powder, which makes the plot a candidate for sorting new powders quickly. For now it is a grouping rather than a validated diagnostic, because we have not independently measured particle size, particle shape, or bulk density.

The two ends of the range are consistent with particle shape. Gas-atomized AlSi10Mg consists of near-spherical particles in a narrow size range, the properties laser powder-bed fusion requires, and it was the fastest powder tested. The fine silicon is milled, angular, and smaller than 44 μm (it passes a 325-mesh sieve). It did not move at all at 0° or 22.5°, and at 45° it produced a single 7 mg pulse in six revolutions. The coarse silicon, the same material at roughly 75–150 μm , delivered 211 mg per revolution. Forces that make particles stick together matter more relative to their weight as particles get smaller, which fits this sharp difference between the two silicon grades. We read the three powders that did not move as a limit of this auger geometry rather than a fixed property of the powders; a wider flight, a larger pitch, or a taller fin would move that limit. The failures also differ in kind. Fine silicon moved slightly at the steepest tilt, so for it the driving force was too weak to overcome cohesion. Fumed silica, which is extremely light and fluffy, never delivered more than a few milligrams per revolution at any tilt and varied wildly from one revolution to the next, which points instead to trapped air.

2.4 Tilt and tapping

Tilt is the strongest setting the controller has. For the nine powders measured at all three angles, raising the tube from horizontal to 22.5° increased delivery by a median factor of 4.2, and raising it further to 45° added a median factor of 1.2 (Fig. 4a). The cohesive powders depended most on gravity: sodium alginate delivered 15 times more per revolution at 45° than horizontally and white rice flour 10 times more, against about 5 times for calcium lactate and sodium sulfate. Tilt was also the only setting that moved one of the non-conveying powders at all (fine silicon, at 45°).

With the auger stopped, nothing flowed on its own. In 33 fifteen-second holds, covering all three tilts on the 11 runs recorded up to 12 August, the balance reading changed by 0.0 mg every time (protocol B). From 20 August, hold readings scattered by tens of milligrams in both directions, including at 0° where gravity cannot empty the tube, so they reflect the bench rather than the powder. Parking the tube horizontally therefore gives a clean shut-off.

A single solenoid tap released between 0.2 mg (sodium alginate) and 20 mg (calcium lactate), or 0.6–10% of one revolution's delivery (Fig. 4b). The size of a tap does not simply follow throughput. Barium chloride and calcium lactate delivered similar amounts per revolution (185 and 198 mg) but released 2.1 and 20 mg per tap, a tenfold difference. Tap increments could be resolved only on the quieter runs; on most fume-hood runs they were smaller than the bench noise.

The vibration motor could not be characterised: its driver

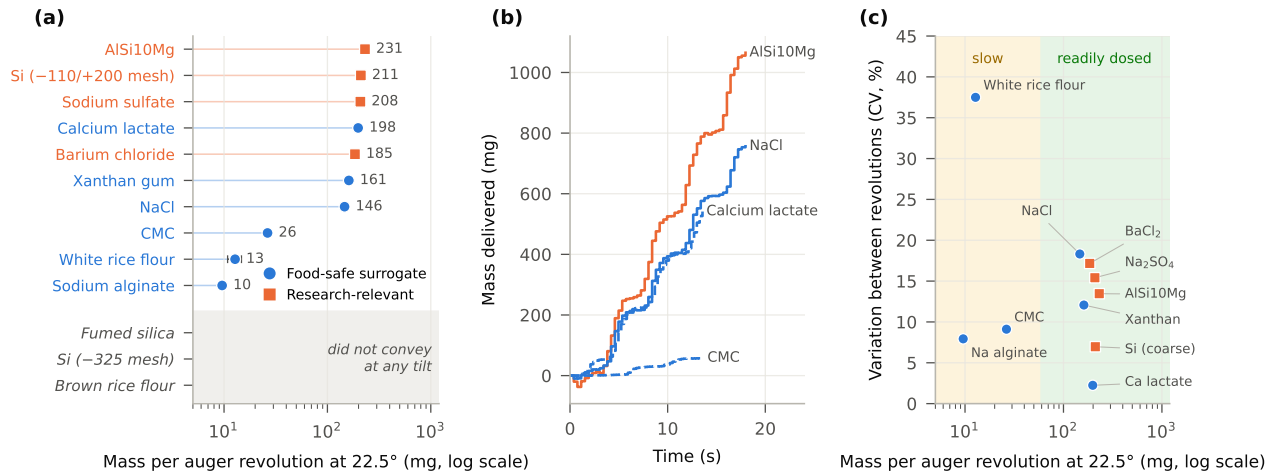


Fig. 3 How much powder the auger delivers, for 13 powders under one fixed set of settings. (a) Mass delivered per auger revolution at 22.5° tilt (protocol C: six revolutions at 30 rpm; error bars show the standard error). The three powders at the bottom delivered nothing distinguishable from zero at any tilt and are listed without a value. (b) Balance readings streamed while the auger turned continuously at 15 rpm and 22.5° tilt (protocol D), for four powders. The mass rises in steps about 4 s apart, one per auger revolution. (c) Mass per revolution against the variation between successive revolutions (coefficient of variation of six revolutions). Both numbers come from protocol C alone; the shading marks the two groups discussed in the text. Blue circles are food-safe surrogates and orange squares research-relevant powders; one run per powder is shown (see Experimental). Fill level was not controlled, particle size and shape were not measured, and all research-relevant powders were tested in the noisier period. Rotations and speeds refer to the auger, and angles to the tube's tilt above horizontal.

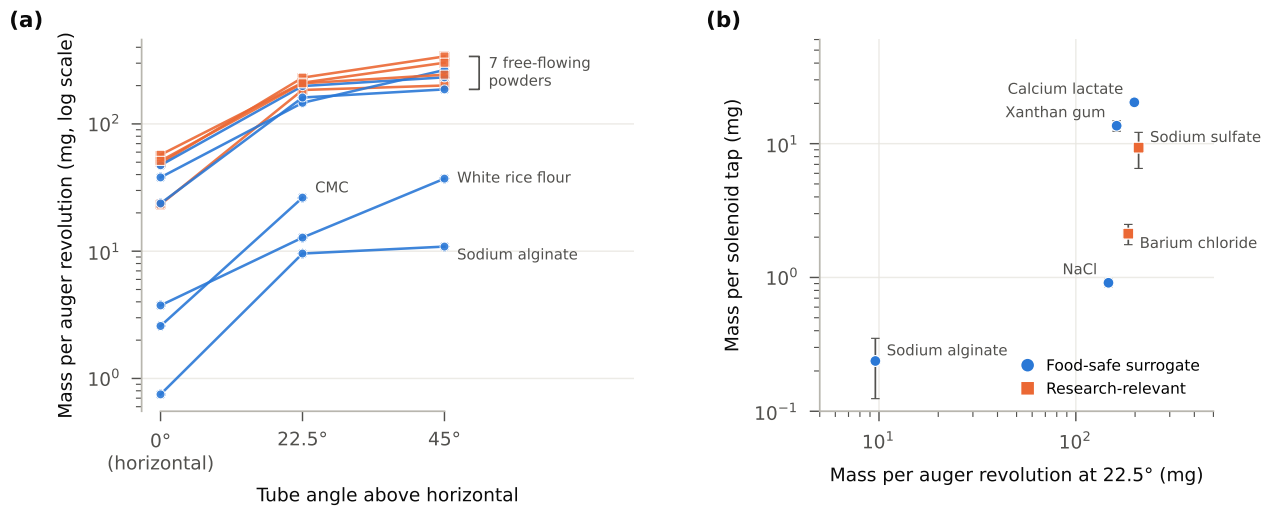


Fig. 4 Effect of tilt and tapping. (a) Mass per auger revolution with the tube horizontal and tilted to 22.5° and 45° (protocol C) for the ten powders that moved. Points are shown only where the mean exceeds twice its standard error, which CMC at 45° did not. The seven free-flowing powders overlap at the top and are bracketed; Table 2 gives their values. On the logarithmic axis, equal vertical distances are equal ratios. (b) Mass released by one solenoid tap (protocol E; mean and standard error of eight taps at 22.5°) against mass per auger revolution, for the six powders whose tap increment exceeded twice its standard error. Markers as in Fig. 3.

Table 2 The 13 powders tested. Mass per auger revolution is the protocol C mean at 22.5° tilt (Fig. 3a). Closed-loop columns give the number of valid doses that finished within the acceptance limit ($\pm 10\%$ at 50 mg, $\pm 5\%$ at 200 mg and 1 g) out of the number run (Fig. 5); a dash means not run. The silicon grades are named by sieve size: -325 mesh passes a $44\ \mu\text{m}$ sieve, and $-110/+200$ mesh lies between about 75 and $150\ \mu\text{m}$.

Powder	Group	mg per rev.	50 mg	200 mg	1 g	Observed behaviour
AlSi10Mg (gas-atomized)	Research	231	1/3	3/3	3/3	Fastest powder; fluid-like; overshoots small targets
Silicon, $-110/+200$ mesh	Research	211	2/2	2/3	3/3	Free-flowing
Sodium sulfate	Research	208	0/3	2/3	3/3	Free-flowing; overshoots small targets
Calcium lactate	Surrogate	198	1/2	2/3	3/3	Free-flowing; largest tap increment
Barium chloride	Research	185	–	–	–	Absorbed moisture and caked in the tube in round 2
Xanthan gum	Surrogate	161	2/3	3/3	3/3	Flows under rotation; can clog after bursts
Sodium chloride (control)	Surrogate	146	2/3	2/3	6/6	Free-flowing baseline
Carboxymethyl cellulose	Surrogate	26	3/3	1/3	3/3	Slow; charges electrostatically
White rice flour	Surrogate	13	3/3	3/3	0/3	Slow and pulsing; most variable between revolutions
Sodium alginate	Surrogate	10	0/3	0/3	0/3	Slowest powder that moved
Silicon, -325 mesh	Research	none	0/3	0/3	0/6	Did not move at 0 or 22.5°; one pulse at 45°
Brown rice flour	Surrogate	none	–	–	0/3	Did not move at any tilt
Fumed silica	Research	none	–	–	–	Did not move; very light and fluffy

Table 3 Test protocols. Every powder went through the same protocols in the same order with frozen settings, so results can be compared across powders; the sequence characterizes the powders and does not tune anything. Rotations and speeds refer to the auger; tilts are the tube's angle above horizontal, from 0° (horizontal park) to 45° (maximum). A trial is one recorded measurement.

Protocol	Varied	Fixed settings	What it measures	Trials
A Balance baseline	Nothing	Tilt 22.5°; eight readings, no actuation	Noise floor of the run, against which every other protocol is read	8
B Static hold	Tilt (0, 22.5, 45°)	15 s hold at each tilt, no actuation	Whether any powder flows on its own (clean shut-off)	3
C Rotation	Tilt (0, 22.5, 45°)	Six single revolutions per tilt at 30 rpm, with a settled reading after each	Mass per revolution and its revolution-to-revolution variation	18
D Speed sweep	Speed (15, 45, 90 rpm)	Tilt 22.5°; three revolutions commanded per speed; balance streamed every 0.25 s	Delivery rate against speed; mass-against-time traces	3
E Tap	Tilt (0, 22.5°)	Eight trials of a one-revolution re-feed followed by one 60 ms solenoid tap	Mass released by a single tap, with the re-feed logged separately	32
F Vibration	Tilt (0, 22.5°)	As E, with three vibration bursts in place of the tap	Mass released by vibration (no data: driver fault)	0
G Closed-loop, 1 g	Nothing	Three doses of 1.000 g; three-phase controller	Dose error, time, and effort per phase	3 doses
H Closed-loop, small	Target (50, 200 mg)	Three doses at each target; same controller	Dose error and time at small targets	6 doses

(DRV2605L) returned an input/output error in every run that asked for protocol F, so the doser as tested relies on rotation, tilt, and tapping alone. The speed sweep (protocol D) suggests that delivery rate grows more slowly than speed, with a median exponent of 0.78 across ten powders for $\text{rate} \propto \text{speed}^a$ over 15–90 rpm (above 1 only for white rice flour, CMC, and sodium alginate). Each speed was measured once, in a fixed order, without re-zeroing the balance, and a firmware timing error let the auger turn 15–54% further than commanded, so we treat these exponents as an indication only and do not plot them.

2.5 Closed-loop dosing

The controller doses in three phases: continuous rotation until the cup is within 0.5 g of the requested mass, then 45° rotations with a settled balance reading after each, and finally pairs of solenoid taps until the reading is within ± 5 mg (see Experimental). All doses used one set of phase settings, tuned by hand on salt and frozen for every powder. In the first test round we ran three doses of 1 g (protocol G) on each powder whose bench conditions allowed it. A second round, from 3 to 15 September 2026, added

three doses each of 50 mg and 200 mg (protocol H) and the 1 g doses that bench noise had prevented for the research-relevant powders. Doses are scored against acceptance limits fixed before testing: $\pm 10\%$ of the requested mass below 100 mg and $\pm 5\%$ at or above it. These limits were chosen so that dosing error adds less than one atomic percent of composition error to a typical five-component alloy blend.^{11,12} Failed doses stay in every count.

At 1 g, every dose of the seven free-flowing and moderately slow powders finished within the $\pm 5\%$ limit, 24 of 24 across salt, calcium lactate, xanthan gum, CMC, AlSi10Mg, coarse silicon, and sodium sulfate (Fig. 5c). The three research-relevant powders were the most accurate: all nine of their 1 g doses finished within 10 mg (1%) of the target, with a median time of 111 s. The four powders that failed at 1 g failed for two different reasons. Fine silicon and brown rice flour delivered nothing, as they had in the open-loop tests. White rice flour and sodium alginate did move, but too slowly: they used up the controller's budget of 200 small rotations and stopped 14% and 29% short of the target after about 14 minutes.

Smaller doses were harder, in different ways for fast and slow powders (Fig. 5a,b). At 200 mg, 18 of 30 doses finished within

$\pm 5\%$, including all three for AlSi10Mg, xanthan gum, and white rice flour. At 50 mg, 14 of 28 finished within $\pm 10\%$. The fast powders now overshoot: of the 12 doses on free-flowing powders that missed the limit at 50 or 200 mg, 11 were overshoots, including all three sodium sulfate doses at 50 mg. For these powders the smallest rotation the controller takes already delivers a large fraction of a 50 mg dose, and delivery arrives in pulses (Fig. 3b), so the last increment can overshoot before the balance reading catches up. Slow powders undershot instead, for the same reason as at 1 g. Overshoot cannot be undone in powder dosing, so for small targets a finer final increment matters more than a faster one.

Time was the main cost of accuracy (Fig. 5d). Doses that finished within the limit took a median of 53 s at 50 mg, 152 s at 200 mg and 185 s at 1 g, far longer than the 30 s we had aimed for. Slow powders took much longer: white rice flour needed 11–13 minutes for each 200 mg dose. The fastest “doses” were failures: fine silicon and brown rice flour stopped after 6–52 s, when the controller detected that nothing was arriving.

Two features of the controller explain most of the remaining failures. First, it never returns to an earlier phase. Once the tap phase begins, it cannot rotate the auger to refill the outlet, apart from 5° nudges that proved too small for the slower powders; when the flight nearest the outlet runs empty, taps alone cannot finish the dose. This is how the first-round 1 g doses of calcium lactate, xanthan gum, and CMC ended: each came to rest 15–49 mg short after 66–166 taps, inside the acceptance limit but outside the controller’s own stopping band. Second, that stopping band was ± 5 mg at every target. At 1 g it is ten times tighter than the ± 50 mg acceptance limit, so the firmware logged these doses as stalled although they met the specification. We report every dose against the acceptance limit and keep the firmware’s own label in the dose record (SI Table S3). A controller that can return to rotation when taps stop working, and whose stopping band scales with the target, is the most direct improvement.

2.6 Limits of the current design

Several limits follow from these results. First, the design covers doses from tens of milligrams to grams. Doses of 0.1–10 mg, which some chemistry applications need, are out of reach for now; they would require per-powder calibration, better isolation from vibration, and a smaller auger. Second, three of the 13 powders did not move, so the current geometry sets a floor on how cohesive or how light a powder can be. Third, the outcome of a dose depends on where the powder happens to sit in the flight when the dose starts. There is no defined starting state, and in routine use the auger would never be reset between doses anyway, so this variability is part of the real performance. Fourth, all masses are what landed in the cup, not what left the tube; powder that missed the cup is not counted, and the tube’s fill level was not controlled. Finally, the balance sets the noise floor. The later runs were made on a balance whose idle readings drifted by up to tens of milligrams; every dose run first had to pass a stability check, but the smallest doses remain the most exposed to this noise.

2.7 Cost and accessibility

The single-channel module (printed parts, motors, drivers, and microcontroller) costs under \$200 in single quantities, excluding the balance (SI Table S1), and prints in under 24 h on a consumer printer. Commercial automated gravimetric dosing modules are quoted at \$30k–\$300k depending on configuration. Even where commercial instruments are more accurate, a cost gap of two orders of magnitude changes what a laboratory can do. It can dedicate one printed channel to each powder so that powders never mix, throw away and reprint the parts the powder touches, and replace a failed channel within a day. This is the economic argument for open-source laboratory hardware^{24,25} applied to the solid-handling gap in SDLs.

3 Experimental

3.1 Fabrication

All printed parts were made in PLA on a Bambu Lab H2D printer (0.2 mm layers, four perimeters, 25% infill), most of them with support material. The auger used in every test is a single printed piece modelled by the team in Fusion 360: a tube 250 mm long with 25 mm outer diameter and 21 mm bore; an 8 mm central core and a single-start flight 0.5 mm thick with 10.4 mm pitch, both confined to the 83 mm nearest the outlet; a tapered outlet with a 3 mm exit hole; a 44-tooth gear (module 1) printed on the outside of the tube 78–88 mm from the outlet; and an external thread at the upper end for the screw-on cap through which the tube is filled. These dimensions were measured from the part file (Fig. 1c). A 20-tooth pinion on the stepper drives the 44-tooth gear (2.2:1 reduction), and the servos drive the tilt plate through a 2:1 gear. The auger, its cap, and the stepper pinion were each modelled in a narrated recording (<https://youtu.be/sKrvGBTUc8U>, <https://youtu.be/EID3ppxmE1U> and <https://youtu.be/E09qnYssKRQ>; repository issues #117 and #120). The earlier AI-modelled auger in the repository instead had a flight along its whole length, four loading slots in the tube wall, and a 16:48 gear pair (3:1). The AI-modelled auger, gears, bracket, tap collar, mounting plate, hinged baseplate, and servo gear sector are parametric models; their source files (build123d, OpenSCAD, and KCL), printable STL files, and renders for every iteration are in the repository. Purchased parts: a NEMA-11 stepper motor (auger drive), two MG996R-class servos (tilt), a 12 V push–pull solenoid (tapping), a coin-type ERM motor (vibration; unused, see Section 2.4), and an A&D HR-100A analytical balance (0.1 mg readability). The auger turns in printed collars rather than bearings.

3.2 Electronics and firmware

A Raspberry Pi Pico W runs MicroPython firmware with commands to rotate by a given angle, tap, vibrate, tilt to an angle, and read the mass. A Pololu Tic T500 drives the stepper over a serial (UART) link, a DRV8871 H-bridge drives the solenoid, and a DRV2605L haptic driver runs the vibration motor over I²C; the balance is read over RS-232 through an SP3232 transceiver board. The electronics began on a breadboard and now sit on a custom printed circuit board (PCB). Schematics, board files, the

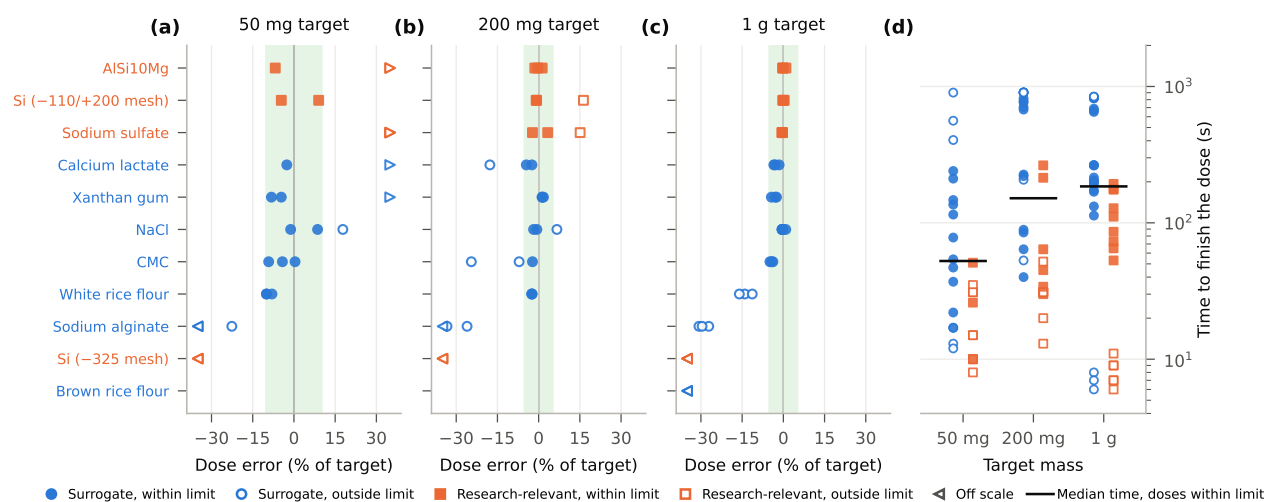


Fig. 5 Closed-loop dosing with one frozen set of controller settings. (a–c) Error of each valid dose at 50 mg, 200 mg and 1 g targets, as a percentage of the target. Green bands mark the acceptance limits ($\pm 10\%$ below 100 mg, $\pm 5\%$ at or above). Filled markers finished within the limit and open markers outside it; triangles at the edge are off scale (the powders that delivered little or nothing). Replicates share a row and are not jittered. Powders are ordered by mass per revolution (Fig. 3a). Doses at 1 g for the food-safe powders are from the first test round (protocol G, August 2026); all 50 and 200 mg doses and the 1 g doses for AISi10Mg, coarse silicon, and sodium sulfate are from the second round (protocols G and H, September 2026). Barium chloride has no valid closed-loop dose (it caked in the tube) and fumed silica was not dosed. (d) Time to finish each dose; horizontal bars are medians of the doses within the limit.

wiring guide, and the firmware are in the repository ([hardware/](#)).

The dosing firmware includes both gear ratios, so its rotation commands and records are in auger revolutions, auger rpm, and tube degrees. The one exception is the test battery, whose raw records store the tilt as a servo setting rather than an angle; the 2:1 tilt gear halves it, so the battery's three tilts are tube angles of 0° , 22.5° , and 45° , and its steepest setting is the 45° maximum. This paper reports tube angles throughout; the conversion is made once, in the analysis scripts, and SI Section S6 maps the raw settings to angles. The auger ratio is confirmed independently by the protocol-D traces, whose pulses recur once per commanded auger revolution (3.94 against 4.00 s).

3.3 Powders and test protocols

Table 2 lists the 13 powders. Each run began by filling the auger tube through its capped upper end; the fill level was not controlled and the loaded tube was not weighed. The test protocols (Table 3) ran in a fixed order with frozen settings, identically for every powder. Each emitted machine-readable trial records over the serial link, which were archived as versioned CSV files stamped with the powder, firmware version, and settings, so every number in this paper can be traced to its raw balance record. A trial is one recorded measurement: in protocol C one trial is a single auger revolution, so its 18 trials are six revolutions at each of three tilts. Protocol D streamed the balance every 0.25 s nominally (0.29–0.39 s in practice) while the auger turned continuously. Its timing loop advanced by the nominal poll interval rather than the elapsed time, so the auger turned 15–54% further than the three revolutions commanded per speed; masses and rates from protocol D use the elapsed time.

Each run was screened for quality before analysis; for example,

a run with a failed tilt servo, a deliberate stress test of the bench, and runs in which the balance could not be read were excluded. One run per powder is shown in Figs. 3 and 4: the newest run that passed screening, preferring runs whose protocols C and E agreed. A mean is reported as a measurement only if it exceeds twice its standard error; otherwise the powder is reported as not having moved. Fumed silica is the one exception to screening: its only run failed it because the outlet could not be seen on camera to rule out a blockage, so its result (an average of at most 0.25 mg per revolution over 350 revolutions) is reported but is less certain than the others. The noise floor of each run is the spread of its eight protocol-A readings.

3.4 Closed-loop dosing procedure

The controller has three phases, with settings that were the same for every powder and every dose. In the bulk phase it tilts the tube to 45° and turns the auger continuously at 55 rpm, stopping 0.12 g early to allow for powder still in flight, until less than 0.5 g remains. In the fine phase it tilts the tube to 22.5° and turns the auger in 45° steps, waiting 1.5 s for a settled reading after each, until less than 50 mg remains. In the tap phase it lowers the tube to horizontal and fires two 60 ms solenoid taps per cycle, again reading the balance after each cycle, until the reading is within ± 5 mg of the target. If three tap cycles in a row add no mass, it turns the auger forward 5° to refill the outlet, up to ten times. For the smaller protocol-H doses, the hand-over points were capped at the target mass (bulk) and at half the target (fine), so a 50 mg dose skips the bulk phase and hands over to tapping with 25 mg left. A dose ends when the reading is within the ± 5 mg band (“ok”), when it exceeds the target by more than the band (overshoot), when tapping and

nudging add nothing more (stalled), after 200 cycles of one phase (cycle budget), or after 900 s. The settings were tuned by hand on salt and then frozen; no dose outcome was used to retune them. In the first round the controller used a reading only once the balance flagged it as stable. After the move to a second fume hood, the balance rarely reported stability while the rig was running, so the second round averaged a bracket of five readings taken 0.25 s apart instead. Before each set of doses, the balance had to stay within the dose tolerance while nothing moved; runs that could not pass this check were postponed rather than run. The reporting follows the gravimetric structure that ISO 8655-6 prescribes for piston pipettes, as adapted for an open-hardware pipette:^{38,39} replicate doses at several target masses, each summarised by its systematic error (mean deviation from target), random error (standard deviation), overshoot rate, and time. Every dose, including failures and the firmware's own termination label, is listed in the dose record (SI Table S3).

LLM coding agents (GitHub Copilot coding agent; Claude-family models; 2026 sessions) carried out design tasks written as GitHub issues. Each session produced a pull request containing the generated CAD code, rendered orthographic and isometric views, and STL files, which the team reviewed in the browser. Literature and design-review queries were run through Edison Scientific. We explored CADSmith and Zoo Design Studio on the same part specifications; in Zoo Design Studio, modelling was done through its Zookeeper agent. No conventional CAD software was used in first designing the parts; the Fusion 360 recreations described in the Results came afterwards, and the tested rig includes some of them (see Fabrication). In line with *Digital Discovery*'s guidance on the use of LLMs, the complete prompt and response history (issues, pull-request threads, agent session logs, and Zoo Design Studio transcripts), model identifiers, and dates are public in the repository and summarised in the SI. Over the project, team members addressed the two coding agents 754 times, and the agents authored 90% of all commits, including those for firmware, analysis, and writing (SI Fig. S3).

A single-channel powder doser with closed-loop weighing can be built from printed parts and commodity electronics for under \$200 plus a balance. With one fixed set of settings, it delivered 10–231 mg per auger revolution for ten of 13 powders and could not move the other three. In closed loop, all 24 doses of 1 g on seven powders finished within $\pm 5\%$ of the target, but doses of 50 and 200 mg met their limits only half to 60% of the time, and a typical dose took minutes rather than seconds. Its main limits are an auger geometry that cannot move very fine or very light powders, a controller that cannot recover once the outlet runs empty, and a balance whose noise sets the floor for small doses.

behind a complete written record of every decision. The 128-entry design log, failures included, is itself a dataset for studying AI-assisted hardware design.

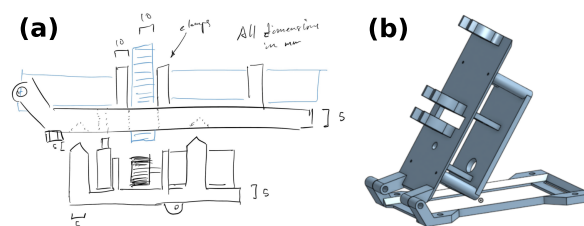


Fig. 6 Future work: the multi-powder doser, in which passive auger cartridges ride on a roller chain past one fixed station that turns, taps, and weighs them (repository issue #128). (a) Hand sketch of the auger carriage, drawn before modelling (dimensions in mm). (b) The auger and frame carriages of the first printed prototype, modelled by the team in Onshape, a conventional CAD package.

S.C.: conceptualization, methodology, investigation, CAD review, writing—original draft. W.M.: investigation, CAD review, validation. L.W.: investigation (electronics, PCB workflows), software. C.R.: investigation, hardware fabrication. G.E.: investigation, hardware fabrication. S.G.B.: conceptualization, supervision, funding acquisition, writing—review & editing.

Conflicts of interest

S.G.B. is a member of the *Digital Discovery* advisory board. There are no other conflicts to declare.

Data availability

All design files (parametric CAD source, the team's Fusion 360 parts as STEP and STL files, and renders), KiCad schematics, MicroPython firmware, AI prompt and response logs, the 128-entry design log, the raw test-protocol records, and the analysis scripts are openly available at <https://github.com/vertical-cloud-lab/powder-doser>. The tidy datasets and scripts behind Figs. 3–5 are in [paper/figures/](#). Hardware designs are released under an open license (see the repository). The datasets will be deposited with a DOI on acceptance. References cited in the SI are listed in this article's reference list.

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Notes and references

- 1 M. Abolhasani and E. Kumacheva, *Nature Synthesis*, 2023, **2**, 483–492.
- 2 G. Tom, S. P. Schmid, S. G. Baird, Y. Cao, K. Darvish, H. Hao, S. Lo, S. Pablo-García, E. M. Rajaonson, M. Skreta, N. Yoshikawa, S. Corapi, G. D. Akkoc, F. Strieth-Kalthoff, M. Seifrid and A. Aspuru-Guzik, *Chemical Reviews*, 2024, **124**, 9633–9732.
- 3 N. J. Szymanski, B. Rendy, Y. Fei, R. E. Kumar, T. He, D. Milsted, M. J. McDermott, M. Gallant, E. D. Cubuk, A. Merchant, H. Kim, A. Jain, C. J. Bartel, K. Persson, Y. Zeng and G. Ceder, *Nature*, 2023, **624**, 86–91.
- 4 M. Seifrid, R. Pollice, A. Aguilar-Granda, Z. M. Chan, K. Hotta, C. T. Ser, J. Vestfrid, T. C. Wu and A. Aspuru-Guzik, *Accounts of Chemical Research*, 2022, **55**, 2454–2466.
- 5 Y. Fei, B. Rendy, R. Kumar, O. Darts, H. P. Sahasrabudde, M. J. McDermott, Z. Wang, N. J. Szymanski, L. N. Walters, D. Milsted, Y. Zeng, A. Jain and G. Ceder, *ArXiv*, 2024.
- 6 A. Aspuru-Guzik, *Digital Discovery*, 2023, **2**, 10–11.
- 7 S. G. Baird, A. R. Falkowski and T. D. Sparks, *ArXiv*, 2026.
- 8 P. M. Maffettone, P. Friederich, S. G. Baird, B. Blaiszik, K. A. Brown, S. I. Campbell, O. A. Cohen, R. L. Davis, I. T. Foster, N. Haghmoradi, M. Hereld, H. Joress, N. Jung, H.-K. Kwon, G. Pizzuto, J. Rintamaki, C. Steinmann, L. Torresi and S. Sun, *Digital Discovery*, 2023, **2**, 1644–1659.
- 9 R. B. Canty, J. A. Bennett, K. A. Brown, T. Buonassisi, S. V. Kalinin, J. R. Kitchin, B. Maruyama, R. G. Moore, J. Schrier, M. Seifrid, S. Sun, T. Vegge and M. Abolhasani, *Nature Communications*, 2025, **16**, 3856.
- 10 Z. Li, A. Ludwig, A. Savan, H. Springer and D. Raabe, *Journal of Materials Research*, 2018, **33**, 3156–3169.
- 11 K. S. Vecchio, O. F. Diplo, K. R. Kaufmann and X. Liu, *Acta Materialia*, 2021, **221**, 117352.
- 12 M. Moorehead, K. Bertsch, M. Niezgoda, C. Parkin, M. Elbakhshwan, K. Sridharan, C. Zhang, D. Thoma and A. Couet, *Materials & Design*, 2020, **187**, 108358.
- 13 S. Mooraj, G. Kim, X. Fan, S. Samuha, Y. Xie, T. Li, J. S. Tiley, Y. Chen, D. Yu, K. An, P. Hosemann, P. K. Liaw, W. Chen and W. Chen, *Communications Materials*, 2024, **5**, 14.
- 14 P. Nelaturu, J. R. Hattrick-Simpers, M. Moorehead, V. Jambur, I. Szlufarska, A. Couet and D. J. Thoma, *Materials Science and Engineering: A*, 2024, **891**, 145945.
- 15 M. Besenhard, E. Faulhammer, S. Fathollahi, G. Reif, V. Calzolari, S. Biserni, A. Ferrari, S. Lawrence, M. Llusa and J. Khinast, *European Journal of Pharmaceutics and Biopharmaceutics*, 2015, **94**, 264–272.
- 16 S. Yang and J. Evans, *Powder Technology*, 2007, **178**, 56–72.
- 17 W. E. Engisch and F. J. Muzzio, *Powder Technology*, 2012, **228**, 395–403.
- 18 W. E. Engisch and F. J. Muzzio, *Powder Technology*, 2015, **283**, 389–400.
- 19 J. Hanson, *Powder Technology*, 2018, **331**, 236–243.
- 20 M. Besenhard, S. Fathollahi, E. Siegmann, E. Slama, E. Faulhammer and J. Khinast, *International Journal of Pharmaceutics*, 2017, **519**, 314–322.
- 21 E. Faulhammer, M. Fink, M. Llusa, S. M. Lawrence, S. Biserni, V. Calzolari and J. G. Khinast, *International Journal of Pharmaceutics*, 2014, **473**, 617–626.
- 22 S. Chianrabutra, B. Mellor and S. Yang, Proceedings of the Solid Freeform Fabrication Symposium, Austin, TX, 2014.
- 23 J. Alsenz, *Powder Technology*, 2011, **209**, 152–157.
- 24 J. M. Pearce, *HardwareX*, 2020, **8**, e00139.
- 25 J. Hohlbein, B. Diederich, B. Marsikova, E. G. Reynaud, S. Holden, W. Jahr, R. Haase and K. Prakash, *Nature Methods*, 2022, **19**, 1020–1025.
- 26 T. Wenzel, *PLOS Biology*, 2023, **21**, e3001931.
- 27 J. Stirling, K. Bumke, J. Collins, V. Dhokia and R. Bowman, *International Journal of Computer Integrated Manufacturing*, 2022, **35**, 1297–1309.
- 28 J. Vasquez, H. Twigg-Smith, J. T. O'Leary and N. Peek, Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, 2020, pp. 1–13.
- 29 B. Subbaraman, O. de Lange, S. Ferguson and N. Peek, *PLOS ONE*, 2024, **19**, e0296717.
- 30 R. Wu, C. Xiao and C. Zheng, *Deepcad: a deep generative network for computer-aided design models*, 2021, <https://doi.org/10.48550/arxiv.2105.09492>, NEEDS MANUAL VERIFICATION: DOI returned non-404 but Crossref metadata unavailable.
- 31 K. D. D. Willis, Y. Pu, J. Luo, H. Chu, T. Du, J. G. Lambourne,

- A. Solar-Lezama and W. Matusik, *ACM Transactions on Graphics*, 2021, **40**, 1–24.
- 32 A. Seff, W. Zhou, N. Richardson and R. P. Adams, *Text*, 2021.
- 33 X. Xu, J. Lambourne, P. Jayaraman, Z. Wang, K. Willis and Y. Furukawa, *ACM Transactions on Graphics*, 2024, **43**, 1–14.
- 34 K. Alrashedy, P. Tambwekar, Z. Zaidi, M. Langwasser, W. Xu and M. Gombolay, *ArXiv*, 2025.
- 35 A. Badagabettu, S. S. Yarlagadda and A. B. Farimani, *ArXiv*, 2024.
- 36 P. A. Jansen, *ArXiv*, 2023.
- 37 T. Rios, S. Menzel and B. Sendhoff, 2023 IEEE Symposium Series on Computational Intelligence (SSCI), 2023, pp. 1704–1711.
- 38 N. Yoshikawa, K. Darvish, M. G. Vakili, A. Garg and A. Aspuru-Guzik, *Digital Discovery*, 2023, **2**, 1745–1751.
- 39 N. Yoshikawa, K. Angers, K. Darvish, S. Okhovatian, D. Bannerman, I. Yakavets, M. Radisic and A. Aspuru-Guzik, *Digital Discovery*, 2026, **5**, 93–97.